EXPERIMENT 21

Syntax-Driven Semantic Analysis

Aim

To perform simple syntax-driven semantic analysis by extracting noun phrases and assigning basic meanings.

Procedure

1. Define a small grammar for noun phrases and sentences.
2. Parse the sentence using NLTK.
3. Extract noun phrase subtrees.
4. Assign a simple semantic label to each noun phrase.
5. Display the extracted phrases and meanings.

Program

```
# Install first: pip install nltk
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the' | 'a'
N -> 'student' | 'book' | 'teacher'
V -> 'reads' | 'helps'
""")
parser = ChartParser(grammar)
sentence = "the student reads a book".split()

for tree in parser.parse(sentence):
    for subtree in tree.subtrees(lambda t: t.label() == "NP"):
        phrase = " ".join(subtree.leaves())
        meaning = "person/entity" if "student" in phrase or "teacher" in phrase else
"object"
        print(phrase, "->", meaning)
    break
```

Output

```
OUTPUT — EXPERIMENT 21
the student -> person/entity
a book -> object
```

Result

Thus, noun phrases were extracted from the parse tree and assigned simple semantic labels.